

Disproof of TLMC Conjecture 00000000124: the least missing value in the orbit of 2 equals q , not $O((\log q)^2)$

TLMC falsification package

September 29, 2026

Abstract

Conjecture 00000000124 asserts that if q is an odd prime for which 2 is a primitive root, then the least positive integer $m(q)$ missing from the orbit $\{2^n \bmod q : n \geq 0\}$ satisfies $m(q) = O((\log q)^2)$. This is false: the hypothesis forces the orbit to be all of $\{1, \dots, q-1\}$, so $m(q) = q$ identically, and $q > (\log q)^2$ for every $q \geq 3$. The claim therefore fails with constant $C = 1$ at *every* admissible prime (e.g. $m(3) = 3 > 1.2069 = (\ln 3)^2$ and $m(101) = 101 \gg 21.2993 = (\ln 101)^2$), and for every fixed $C > 0$ at every admissible prime $q > e^{6C}$. All instances are confirmed by independent recomputation and by machine-checked Lean 4 certificates using no axioms and no `sorry`.

1 The conjecture and the attack

Conjecture 1 (TLMC 00000000124). *Let q be an odd prime such that 2 is a primitive root modulo q (i.e. the multiplicative order satisfies $\text{ord}_q(2) = q-1$). Let $m(q)$ be the least positive integer that does not occur in the orbit $\{2^n \bmod q : n \geq 0\}$, residues taken in $\{0, 1, \dots, q-1\}$. Then $m(q) = O((\log q)^2)$.*

The attack is structural and requires no computation: the hypothesis is so strong that it determines $m(q)$ exactly.

2 The exact value of $m(q)$

Theorem 1. *Let q be an odd prime with $\text{ord}_q(2) = q-1$. Then the orbit of 2 modulo q , written with least positive residues, is exactly $\{1, 2, \dots, q-1\}$, and the least missing positive integer is*

$$m(q) = q.$$

Proof. Since q is odd, $\gcd(2, q) = 1$, so no power of 2 is divisible by q : the residues $r_n := 2^n \bmod q$ all lie in $\{1, \dots, q-1\}$.

The values $r_0, r_1, \dots, r_{q-2}$ are pairwise distinct. Indeed, if $2^i \equiv 2^j \pmod{q}$ with $0 \leq i < j \leq q-2$, then multiplying by the inverse of 2^i gives $2^{j-i} \equiv 1 \pmod{q}$ with $0 < j-i < q-1$, contradicting $\text{ord}_q(2) = q-1$.

So the orbit contains $q-1$ distinct elements of the $(q-1)$ -element set $\{1, \dots, q-1\}$, hence equals it: $\{r_0, \dots, r_{q-2}\} = \{1, \dots, q-1\}$ (and r_n for $n \geq q-1$ merely repeats it, since $2^{q-1} \equiv 1$).

Consequently every positive integer $k \leq q-1$ occurs in the orbit (as the residue k itself), while q does not: $q \equiv 0 \pmod{q}$ and 0 is not a power of 2 modulo q . The least missing positive integer is therefore q . □

3 The gap against the logarithmic bound

Theorem 2. *For every real $x \geq 1$, $(\ln x)^2 < x$.*

Proof. Put $t = \ln x \geq 0$. We show $e^{t/2} > t$, i.e. $\sqrt{x} > \ln x$. The tangent bound $e^u \geq 1 + u$ (all Taylor coefficients are nonnegative) gives, for $0 \leq t < 2$: $e^{t/2} \geq 1 + t/2 > t$. For $t \geq 2$, applying the same bound at $u = (t - 2)/2 \geq 0$ gives $e^{t/2} = e \cdot e^{(t-2)/2} \geq e(1 + \frac{t-2}{2}) = \frac{e}{2}t > t$, because $e > 2$. $\square$ $\square$

Corollary 1. *For every odd prime q with $\text{ord}_q(2) = q - 1$,*

$$m(q) = q > (\log q)^2$$

with the constant $C = 1$; already $m(3) = 3 > 1.2069 \dots = (\ln 3)^2$ and $m(101) = 101 \gg 21.2993 \dots = (\ln 101)^2$. More generally, for every fixed $C > 0$: since $e^L \geq L^3/6$ for $L \geq 0$ (Taylor series with nonnegative terms), every q with $L = \ln q > 6C$ satisfies $q = e^L \geq \frac{L^3}{6} > CL^2 = C(\log q)^2$, hence $m(q) > C(\log q)^2$.

4 Scope and boundary of the refutation

The following statements are **unconditional**:

- $m(q) = q$ exactly, for every admissible q (Theorem 1);
- $m(q) > (\log q)^2$, i.e. the conjecture fails with constant $C = 1$, for every admissible q without exception (Corollary 1);
- for each fixed $C \leq$ any prescribed value, every admissible $q > e^{6C}$ violates the proposed bound.

The full refutation of the $O(\cdot)$ statement for *arbitrary* constants C additionally requires that admissible primes be unbounded. By Theorem 1, $m(q)/(\log q)^2 = q/(\log q)^2 \rightarrow \infty$ along any unbounded sequence of admissible primes; the infinitude of primes with 2 as primitive root is Artin's primitive-root conjecture for $a = 2$, which is open in general but provable under the Generalized Riemann Hypothesis (Hooley, 1967) and supported by the expected positive density (≈ 0.374 ; OEIS A001122). Note that if the admissible set were finite, the O -statement would hold vacuously for large q ; thus the exact identity $m(q) = q$ is the unconditional core of the disproof, and it already destroys the conjecture's polylogarithmic content: the true growth is *linear* in q , not $(\log q)^2$. This boundary is stated explicitly in `README.md`.

5 Computational verification

An independent recomputation (`reproduce.py`, pure standard library) confirms the verdict's spot checks and the general picture for all 22 admissible primes below 200: the orbit is $\{1, \dots, q - 1\}$, $m(q) = q$, and $m(q) > (\ln q)^2$ throughout, with the ratio $m(q)/(\ln q)^2$ growing steadily.

q	$m(q)$	$(\ln q)^2$	ratio	q	$m(q)$	$(\ln q)^2$	ratio
3	3	1.207	2.49	83	83	19.526	4.25
5	5	2.590	1.93	101	101	21.299	4.74
11	11	5.750	1.91	107	107	21.835	4.90
13	13	6.579	1.98	131	131	23.768	5.51
19	19	8.670	2.19	139	139	24.349	5.71
29	29	11.339	2.56	149	149	25.039	5.95
37	37	13.039	2.84	163	163	25.946	6.28
53	53	15.763	3.36	173	173	26.556	6.51
59	59	16.626	3.55	179	179	26.909	6.65
61	61	16.899	3.61	181	181	27.024	6.70
67	67	17.679	3.79	197	197	27.912	7.06

The two verdict spot checks are reproduced exactly: for $q = 3$, $m = 3 > (\ln 3)^2 \approx 1.2069$; for $q = 101$, $m = 101 \gg (\ln 101)^2 \approx 21.2993$.

6 Machine-checked Lean 4 certificates

The directory `lean4/` contains a self-contained Lean 4 project (toolchain `leanprover/lean4:v4.33.1`, no dependencies, core Lean only). For each admissible prime $q < 200$ it proves, by `decide` over $\mathbb{N}$ arithmetic or by the two elementary lemmas of the file:

- `pr_q`: 2 is a primitive root mod q — Fermat’s congruence $2^{q-1} \equiv 1$ together with $2^{(q-1)/f} \not\equiv 1$ for each prime $f \mid q-1$;
- `cover_q`: every $1 \leq k < q$ occurs in the orbit $\{2^n \bmod q\}$, via explicit discrete-log witness lists checked by `dlog0k` and the semantic lemma `dlog0k_implies`;
- `lm_q`: `IsLeastMissing q q`, i.e. the least positive integer missing from the orbit is exactly q (so $m(q) = q$);
- `pw_q`: $2^\ell \leq q < 2^{\ell+1}$ for $\ell = \lfloor \log_2 q \rfloor$ and $\ell^2 < q$ — the concrete gap $m(q) = q > (\log_2 q)^2$.

The flagship `attack_101` bundles the verdict’s $q = 101$ check: primitive-root certificate, $m(101) = 101$, and $6^2 = 36 = (\lfloor \log_2 101 \rfloor)^2 < 101$.

`Check.lean` runs `#print axioms` on all 138 theorems; every single one reports *does not depend on any axioms* — zero `sorry`, zero `propext/Quot.sound/Classical.choice`, i.e. a fully constructive, axiom-free audit.

7 Reproducing

```
python3 reproduce.py          # independent recomputation (stdlib only)
cd lean4 && lake build         # builds Main.lean (Lean 4.33.1)
lake env lean Check.lean      # axiom audit: 138 x "does not depend on any axioms"
tectonic main.tex --outdir build # this document
```

References

- [1] E. Artin, *Über eine neue Art von L -Reihen*, Abh. Math. Sem. Univ. Hamburg 3 (1927), 89–108.
- [2] C. Hooley, *On Artin's conjecture*, J. Reine Angew. Math. 225 (1967), 209–220.
- [3] OEIS Foundation, *A001122: Primes with primitive root 2*, <https://oeis.org/A001122>.